Cross-Domain Remote Sensing Scene Classification for Balancing Transferability and Discriminability in Partial Domain Adaptation

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***Abstract*—Due to the high cost of annotating remote sensing images, unsupervised partial domain adaptation (PDA) that does not require manual annotation has been widely studied for application in remote sensing scene classification (RSSC) tasks. PDA is based on the assumption that the class space of the target domain is a subset of the source domain. Therefore, the model trained on the large-scale and full-labeled source domain is transferred to the small-scale and unlabeled target domain effectively. However, most existing PDA methods for RSSC focus only on improving the transferability of features in the domain alignment process, neglecting the discriminative information that is equally important for cross-domain classification tasks. In this article, we proposed a synergistic contrastive-adversarial learning PDA (SCAPDA) method for adaptive cross-domain RSSC. The SCAPDA is built upon a domain adversarial network, incorporating a discrimination constraint module (DCM) tailored for the target domain and a supervised contrastive learning module (SCLM) operating on both the source and target domains. The DCM learns a target-specific classifier under the constraint of enhancing discriminability in the target domain, thereby providing high-confidence supervision signals for supervised contrastive learning. The SCLM leverages supervised contrastive learning to explore both domain-invariant and class-specific information across source and target samples, aiming to promote transferable representations while preserving discriminability. The effectiveness of the proposed method is validated through experiments on three publicly available remote sensing datasets.**

***Index Terms*—Negative transfer, partial domain adaptation (PDA), remote sensing scene classification (RSSC), supervised contrastive learning.**

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I. Introduction

With the increasing number of satellites and the improvement in the quality of satellite imagery, optical remote sensing has attracted growing attention in various remote sensing understanding and analysis tasks [1], [2], [3], bringing new opportunities for research in environmental monitoring, maritime security, and agricultural management. However, data-driven deep learning and large-scale model techniques are constrained by the scarcity of annotated image data [4], [5], which poses significant challenges for cross-domain remote sensing scene classification (RSSC). To relax the need for abundant labeled data, unsupervised partial domain adaptation (PDA) provides a promising solution by assuming that the label space of the target domain is a subset of that of the source domain. PDA aims to transfer knowledge from a label-rich source domain to an unlabeled target domain. Therefore, training models on source domains with a large number of labeled classes and applying them to target domains with fewer classes offers a more practical and meaningful approach in real-world remote sensing tasks.

In the field of RSSC, PDA has made notable progress in recent studies [6], [7], [8]. On the one hand, benefiting from the emergence of the domain adversarial network [9], some existing methods focus on leveraging adversarial learning to encourage domain confusion. This strategy aims to extract domain-invariant features, thereby promoting positive transfer across domains. On the other hand, some methods aim to suppress negative transfer by reweighting source domain samples or constructing auxiliary domains to filter out outlier classes, ensuring that the alignment between source and target domains is restricted to the shared label space. It can be observed that existing PDA methods for RSSC primarily focus on feature transferability, e.g., the ability of learned features to generalize across different remote sensing scenarios. However, these methods overlook discriminative information, which is equally critical for remote sensing image classification tasks.

For example, as illustrated in Fig. 1(a), "bridge" and "overpass" represent two distinct but visually similar scene classes with different data distributions. The key discriminative information between them tends to lie in scene-level spectral characteristics rather than in the spatial geometric structure of the foreground objects. If such discriminative information is ignored, the model tends to overemphasize feature

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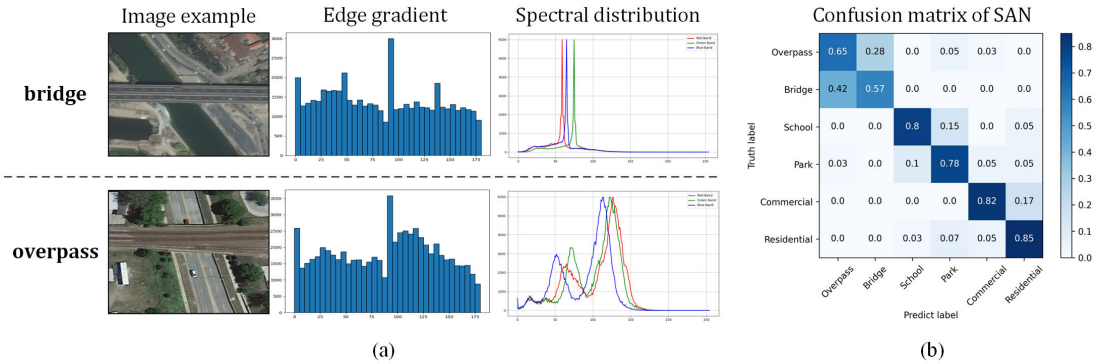


Fig. 1. Illustration and analysis of discriminative representation. (a) Examples of discriminative information in RSSC. The spatial structure is represented by edge gradients, while the spectral characteristics are described by spectral distributions. (b) Confusion matrix results of the SAN on six scene classes.

transferability during training, leading to overfitting and consequently a poorer generalization error bound [10]. Fig. 1(b) presents the confusion matrix of the SAN [11] model, which focuses solely on transferability. In certain scenarios, confusable classes such as "overpass" and "bridge" achieve classification accuracies of around 0.6, whereas other scenes reach about 0.8. Therefore, the performance of the classic PDA network SAN is easily constrained by these confusable classes, as it focuses solely on feature transferability while neglecting discriminability.

Unfortunately, existing studies have demonstrated an inherent conflict between transferability and discriminability under domain adversarial networks [12]. Transferability is mainly dominated by the principal directions associated with the largest singular values, whereas discriminability typically relies on a larger number of fine-grained feature directions corresponding to smaller singular values. However, during adversarial alignment, these small-singular-value directions are often regarded as the primary sources of domain discrepancy and are therefore suppressed [12], which inevitably weakens the discriminative information. Although several PDA methods in the general vision domain [13], [14] attempt to preserve discriminability, they are difficult to effectively apply to complex remote sensing scenarios due to the pronounced interclass similarity and intraclass diversity exhibited by optical remote sensing images. More specifically, remote sensing scene images are characterized by high spatial resolution and rich structural details, resulting in abundant high-frequency spatial information. Consequently, the singular value spectrum of the feature matrix tends to be more sharply distributed. Moreover, domain adversarial training further amplifies the dominant singular directions, thereby exacerbating the suppression of discriminative information carried by smaller singular values. As a result, compared with conventional natural images, adversarial networks exhibit a stronger bias toward high-singular-value directions on remote sensing imagery, making the effective preservation and enhancement of discriminative representations a particularly critical issue.

Inspired by linear discriminant analysis (LDA) [15], which measures feature discriminability via the ratio of interclass variance to intraclass variance—with a larger ratio indicating higher discriminability—we introduce supervised contrastive learning to enhance cross-domain discriminative capability. Supervised contrastive learning explicitly enforces intraclass compactness while enlarging interclass separability, thereby facilitating the formation of more discriminative feature structures in the target domain. Essentially, during backpropagation, supervised contrastive learning strengthens fine-grained directions associated with smaller singular values, effectively preserving the discriminative information encoded in feature vectors corresponding to these small singular values [16].

To ensure feature transferability while preserving discriminative capability, we propose a synergistic contrastive-adversarial learning PDA (SCAPDA) method for adaptive cross-domain RSSC. Built upon a domain adversarial network framework, we introduce a discrimination constraint module (DCM) to enhance intraclass compactness within the target domain, and a supervised contrastive learning module (SCLM) is constructed to collaboratively train with cross-domain adversarial learning. In particular, we aim to learn feature representations by maximizing intraclass similarity and minimizing interclass similarity, and leveraging class information to enhance feature discriminability. First, the DCM explores the intrinsic discriminative structure of the target domain by introducing two auxiliary classifiers to impose discrimination constraints on target features. The pseudolabels of target samples are generated using the target-specific classifier with higher prediction confidence, which helps mitigate the risk posed by noisy predictions. Then, in the SCLM, the outputs of the target-specific classifier are used as supervision signals to mine invariant information among samples from the same class across domains, as well as class-specific features between different classes. Thus, SCAPDA implicitly guides adversarial training toward a better equilibrium and explicitly calibrates decision boundaries. The main contributions of this work are summarized as follows.

1) We propose a SCAPDA framework for RSSC. Unlike existing PDA methods for RSSC, which primarily focus on either feature alignment or outlier class suppression, SCAPDA jointly considers feature transferability and discriminability within a unified model. By leveraging adversarial learning and the SCLM for cross-domain alignment and class-structure learning, SCAPDA alleviates negative transfer and enhances target domain discriminability, enabling more robust knowledge transfer.

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2) We design a DCM for the target domain. DCM employs two auxiliary classifiers to detect hard samples near the target-domain decision boundary and learns a stable discriminative structure through a discrepancy-based constraint. Meanwhile, a target-domain-specific classifier generates high-confidence pseudolabels for SCLM, thereby alleviating pseudolabel error accumulation to a certain extent.
3) We conduct extensive experiments on PDA settings constructed from three public RSSC datasets (AID, NWPU-RESISC45, and UC Merced). The results show that the proposed method consistently outperforms existing approaches across various PDA scenarios, validating its effectiveness in enhancing cross-domain discriminability and mitigating negative transfer.

II. Related Work

This work leverages the inherent mechanism of contrastive learning, which enhances interclass discriminability and preserves intraclass consistency, to jointly coordinate the convergence of adversarial PDA. To contextualize it, we overview recent advances in PDA and contrastive learning.

A. Partial Domain Adaptation

Unlike standard unsupervised domain adaptation (UDA), PDA assumes that the target-domain label space is a subset of the source-domain label space. Misaligning source-domain outlier classes with the target domain can lead to negative transfer. Therefore, most current studies employ adversarial learning for distribution alignment and suppress negative transfer by reducing source-domain outlier sample weights. Cao et al. [11] proposed the selective adversarial network, which employs multiple discriminators for class-wise alignment while suppressing negative transfer from irrelevant source samples. Zhang et al. [17] proposed an adaptive weighting mechanism to dynamically adjust class weights during training. Cao et al. [18] further applied prediction-based weighting to both the discriminator and shared classifier to improve adaptation performance. Based on this, Cao et al. [19] introduced a similarity-based transferability estimation strategy to reweight the discriminator and classifier, thereby improving weight-estimation quality. Li et al. [20] argued that some source-domain features may be unsuitable for the target task and proposed a residual correction learning method to mitigate feature discrepancies between source and target domains. Based on SAN, Cao et al. [21] proposed an improved version named SAN++. Luo and Ren [22] introduced a label-mask-based class distribution alignment method and highlighted insufficient discriminability in PDA. Yang et al. [13] incorporated contrastive learning into adversarial adaptation and proposed a clustering-guided reweighting mechanism to enhance discriminability. Sahoo et al. [14] proposed a three-stage selection–labeling–mixing framework to learn discriminative invariance and mitigate negative transfer. He et al. [23] proposed a self-training strategy with contrastive learning to address overfitting and mine domain-specific information in PDA. Gu et al. [24] introduced a new uncertainty loss for PDA to reduce expected errors on the target domain. In addition, Li and Chen [25] explored nonaligned PDA by preserving manifold structures to facilitate knowledge transfer.

Currently, the application of PDA algorithms in real-world scenarios has been extensively studied [26], [27], [28], [29], [30], with some notable progress achieved in RSSC. Hu et al. [6] proposed a novel coordinate loss for class reweighting. Zheng et al. [7] improved the DANN network [9] by introducing a progressive auxiliary domain module and complementary entropy regularization for remote sensing PDA. Li et al. [8] proposed a selective alignment Transformer with class- and patch-level reweighting to enhance transferability in remote sensing PDA. However, when dealing with complex remote sensing scenes that exhibit high interclass similarity, such methods may suffer from misclassification. In contrast to the aforementioned approaches, our method not only focuses on enhancing the transferability during domain adaptation but also explicitly considers the discriminability of complex target scenes.

B. Contrastive Learning

Although supervised learning achieves strong performance, it relies heavily on costly, labor-intensive, large-scale annotated datasets. In label-scarce scenarios, contrastive learning offers a promising alternative. By leveraging unlabeled data, contrastive learning enables models to learn discriminative representations by comparing similar and dissimilar sample pairs, thereby enhancing generalization. The general paradigm takes an original sample (anchor) x and applies data augmentation techniques, such as rotation, cropping, resizing, or color jittering, to generate a corresponding positive sample x^+. The pair (x, x^+) forms a positive pair, while the remaining samples x^- in the same minibatch are treated as negative samples, forming negative pairs (x, x^-) with the anchor. The learning objective encourages the model to bring positive pairs closer in the representation space while pushing negative pairs farther apart. This process is typically formalized as follows:

$$\text{score}\left(f(x), f\left(x^+\right)\right) \gg \text{score}\left(f(x), f\left(x^-\right)\right) \tag{1}$$

where $f(x)$ denotes the feature extraction function and the score represents a similarity metric quantifying the distance between sample pairs. Through this paradigm, the similarity between semantically related samples is amplified, while the dissimilarity between unrelated samples is emphasized.

To integrate contrastive learning with downstream tasks, including image classification, segmentation, and object detection, various approaches have been proposed, including mask-based methods [31], [32], object-based methods [33], [34], region-based methods [35], [36], and instance-based methods. Among these, instance-based contrastive learning [37], [38], [39], [40] treats augmented views of the same sample as positives and other samples in the batch as negatives, showing promising progress in recent years. For example, Devon Hjelm et al. [40] maximized mutual information between global and local features to incorporate local structural information into learned representations. Tian et al. [41] treated different views of the same scene as positive pairs

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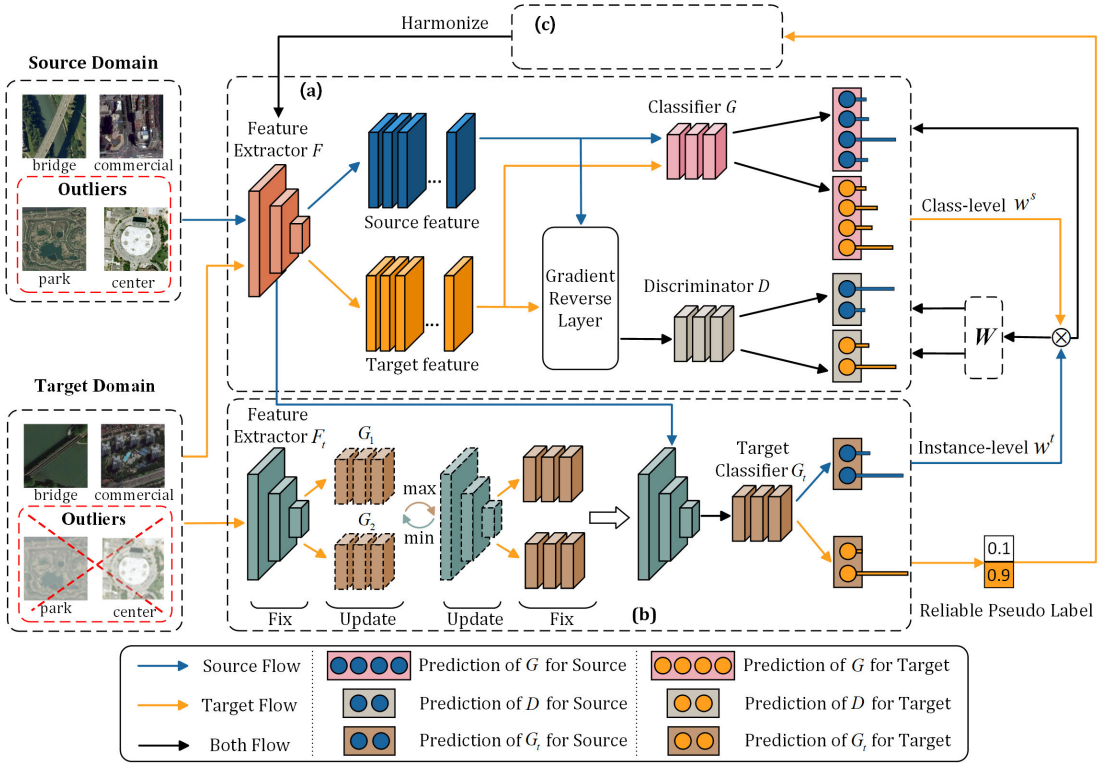


Fig. 2. Overall framework of SCAPDA. In particular, F is the feature extractor of the backbone network, D is the discriminator of the adversarial network, and G is the classifier of the backbone. F_t, G_1, and G_2 are the feature extractor and classifiers used to constrain the discriminability of the target domain. G_t is a domain-specific classifier based on the discriminative features of the target domain. (a) Domain adversarial network. (b) Discrimination constraint. (c) Supervised contrastive learning.

in contrastive learning to maximize mutual information. To address batch-size-limited negative sample capacity, He et al. [42] proposed momentum contrast (MoCo), which uses a dynamic negative sample queue and a momentum encoder to maintain representation consistency. Chen et al. [43] proposed SimCLR, improving representation quality through diverse data augmentations and a nonlinear projection head.

In addition, clustering-based methods [44], [45], [46] emphasize that same class samples should be closer in the feature space. Caron et al. [45] introduced local view cropping and clustering-based learning to enforce consistency across augmented views. Khosla et al. [46] proposed a supervised contrastive learning framework that uses label information to cluster same class samples in the embedding space. Since self-supervised contrastive learning ignores class information in sample-pair construction, we adopt supervised contrastive learning with reliable pseudolabels to exploit class priors and balance transferability and discriminability.

III. Synergistic Contrastive-Adversarial Learning for PDA

A. Problem Formulation

Similar to UDA, PDA for RSSC involves a labeled source domain $D_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$ drawn from distribution $p(x)$, and an unlabeled target domain $D_t = \{x_i^t\}_{i=1}^{n_t}$ drawn from a different distribution $q(x)$. The source and target domains contain C_s and C_t scene classes, respectively, where n_s and n_t denote the sample numbers in each domain. The key difference lies in the label space: the target domain shares only a subset of source-domain classes, i.e., $C_t \subseteq C_s$. In other words, C_t represents the shared classes, while the source domain contains additional private classes, denoted as $\overline{C_s}$. To illustrate the challenges of PDA, we compare it with semi-supervised learning (SSL). SSL typically assumes that source and target data are drawn from the same distribution and share an identical label space. On the contrary, PDA involves both distribution and label-space shifts. The domain distributions and label-space settings of PDA, UDA, and SSL are summarized in Table I. The unique challenge of PDA arises from handling domain shift $p(x) \neq q(x)$ together with label-space shift $C_t \subseteq C_s$, making it more complex than both UDA and SSL.

TABLE I
Different Task Settings of SSL, UDA, and PDA

Paradigm	Domain distribution	Class space set
SSL	$p(x) = q(x)$	$C_s = C_t$
UDA	$p(x) \neq q(x)$	$C_s = C_t$
PDA	$p(x) \neq q(x)$	$C_s \supseteq C_t$

Therefore, PDA poses two main challenges: 1) filtering out source private classes to mitigate negative transfer; 2) reducing domain shift, i.e., $p(x) \neq q(x)$, to facilitate positive transfer. In addition, given the complex spatial structures of remote sensing images, we further investigate how to enhance both feature transferability and discriminability for more effective domain adaptation.

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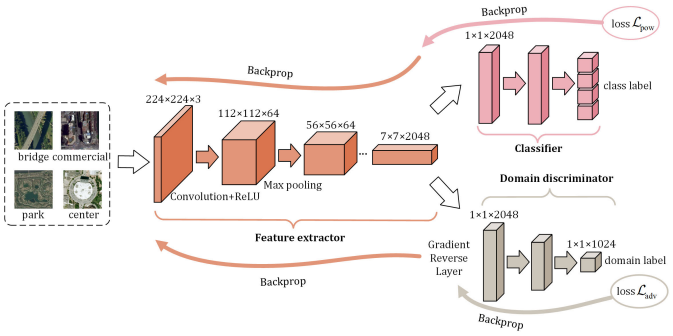


Fig. 3. Structure of domain adversarial network.

B. Overall Framework

To address performance degradation caused by existing PDA methods for RSSC neglecting model discriminability during training, we propose a SCAPDA approach. The overall training framework, illustrated in Fig. 2, consists of three main components: (a) domain adversarial network, (b) DCM, and (c) supervised contrastive learning model. The adversarial network includes a pretrained feature extractor F and a domain discriminator D. Through adversarial training, the two converge, enabling F to extract domain-invariant features for training a transferable classifier G. The DCM contains a feature extractor F_t focused on the target domain, two auxiliary classifiers G_1 and G_2, and a target-specific classifier G_t. The two auxiliary classifiers impose discrimination constraints on the target feature space to obtain a highly discriminative classifier G_t and provide reliable supervision signals for contrastive learning. Supervised contrastive learning is applied simultaneously in the source and target domains to extract richer discriminative information. Finally, the classifiers G and G_t jointly generate weight feedback to the adversarial network, filtering out irrelevant source-domain classes.

C. Domain Adversarial Network

Recent studies [13], [21], [47] have shown that domain adversarial networks can effectively alleviate domain shift by extracting domain-invariant features. The detailed architecture of the domain adversarial network is shown in Fig. 3, consisting of three parts: a feature extractor F, a domain discriminator D, and a classifier G. The feature extractor F, a convolutional neural network, maps input data into a specific feature space. The domain discriminator D, composed of three fully connected layers, distinguishes whether the input features come from the source or target domain and generates corresponding domain labels. The classifier G serves as the classification head to produce class predictions for optimizing the cross-entropy loss. To obtain domain-invariant representations from both domains for effective transfer, F and D are optimized in a min–max game using adversarial loss, with a gradient reversal layer applied during backpropagation to enable adversarial optimization. As a result, F learns transferable domain-invariant features, allowing the source-trained classifier G to adapt effectively to the target domain.

Since the source and target domains do not share the same class space, for the outlier classes $\overline{C}_s$, the discriminator receives only source-domain samples, providing uninformative signals for domain adaptation. These misleading signals may adversely affect the feature extractor F, thereby reducing its ability to capture transferable features. Consequently, strict domain adversarial alignment can degrade model performance and cause negative transfer. A mainstream solution is to introduce the transferability probability of each source-domain class and reduce the weights of irrelevant classes by reweighting source-domain samples [18], [19], [21]. In this way, alignment is restricted to the shared class set C_t, encouraging F to learn truly domain-invariant features. The reweighted adversarial loss $\mathcal{L}_{adv}$ is defined as follows:

$$\mathcal{L}_{adv} = \frac{1}{n_s}\sum_{i=1}^{n_s} w_i \log\left[D\left(F\left(x_i^s\right)\right)\right] + \frac{1}{n_t}\sum_{j=1}^{n_t} \log\left[1 - D\left(F\left(x_j^t\right)\right)\right] \tag{2}$$

where w_i denotes the weight assigned to each source sample. Lower weight samples receive less emphasis during training, thereby filtering out outlier classes. The detailed formulation is introduced in Section III-E. The adversarial network incorporates target-data prediction uncertainty, quantified by maximizing $\mathcal{H}_\alpha(\mathbf{p}) = \sum_k p_k^\alpha$. When $\mathcal{H}_\alpha(\mathbf{p})$ reaches its maximum, the probability distribution $\mathbf{p}$ approaches a one-hot vector, indicating minimal prediction uncertainty. The prediction-uncertainty-based classification optimization loss $\mathcal{L}_{pow}$ is defined as follows:

$$\mathcal{L}_{pow}\left(\theta_f, \theta_g\right) = \frac{1}{n_s}\sum_{i=1}^{n_s} w_i l_{ce} G\left(F\left(x_i^s\right)\right) - \frac{1}{n_t}\sum_{j=1}^{n_t} \mathcal{H}_\alpha\left(G\left(F\left(x_j^t\right)\right)\right) \tag{3}$$

where x^s and x^t denote source and target domain samples, respectively. $\mathbf{w} = \{w_1, w_2, \ldots, w_{n_s}\}$ represents the weights for each source-domain class. l_{ce} denotes the predefined cross-entropy loss function.

D. Discrimination Constraint Module

In the PDA task, features can be categorized into two types based on task characteristics: transferable features and discriminative features. According to [48], transferable features are further divided into domain-invariant and class-invariant types, while discriminative features are categorized into domain-specific and class-specific types. Among these, we focus primarily on domain-invariant features and class-specific features—the former facilitates domain alignment, while the latter contributes to classification performance. Although a tradeoff exists between the two, both are not always simultaneously necessary; some specific tasks may rely predominantly on just one. Based on this insight, we design a target domain-specific classifier fully constrained by discriminative features (i.e., class-specific features). This classifier first learns the discriminative feature distribution of the target domain and then is trained accordingly. Since the classifier is dedicated exclusively to the target domain and

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fully exploits the discriminative structure of target samples, it is capable of producing highly reliable class labels for the target domain, which in turn guide the subsequent supervised contrastive learning.

To capture the discriminative feature distribution of the target domain, we leverage prediction discrepancies among auxiliary classifiers to identify hard samples, i.e., target samples located near or beyond the source-domain decision boundary. Conversely, simple samples are target samples correctly predicted by the auxiliary classifiers. Subsequently, through adversarial optimization between the auxiliary classifiers and the feature extractor, these boundary samples are compacted into the correct class boundaries, thereby revealing a distinct discriminative structure of the target domain. Since the core idea relies on prediction discrepancies among classifiers, which cannot be achieved with a single classifier, multiple auxiliary classifiers are necessary.

Suppose $K(K > 1)$ auxiliary classifiers are introduced, and their prediction set for a given sample is denoted as $\mathbf{S} = \{s_1, s_2, \ldots, s_K\}$. The discrepancy metric is defined as follows:

$$D_K = \frac{2}{K(K-1)} \sum_{1 \le i < j \le K} \left(s_i - s_j\right) \tag{4}$$

where s_i denotes the output of each classifier, modeled as $s_i = \mu + \epsilon_i$, μ is the mean, and ϵ_i represents independent random perturbation noise with $\mathrm{Var}(\epsilon_i) = \sigma^2$. Thus, $\mathrm{Var}(s_i - s_j) = \mathrm{Var}(\epsilon_i - \epsilon_j) = 2\sigma^2$. Therefore, the variance of the averaged discrepancy is given by

$$\begin{aligned} \mathrm{Var}\left(D_K\right) &\approx \left(\frac{2}{K(K-1)}\right)^2 \sum_{1 \le i < j \le K} \mathrm{Var}\left(s_i - s_j\right) \\ &\approx \frac{4\sigma^2}{K(K-1)}. \end{aligned} \tag{5}$$

We further clarify the use of $\approx$ in (5). Although classifiers sharing feature inputs are not strictly independent, this approximation has been widely adopted in prior work [49]. Moreover, to reduce classifier correlation, we initialize them with different random seeds, introducing parameter diversity and partly alleviating output dependency. From this formulation, as the number of auxiliary classifiers K increases, the variance of the discrepancy decreases, implying smaller fluctuations in the averaged discrepancy. This weakens the discrepancy signal and reduces the ability to detect boundary samples. Consequently, we adopt the double-auxiliary-classifier setting to maximally capture discrepancy signals.

In particular, inspired by [50], we employ a double-classifier architecture with two auxiliary classifiers G_1 and G_2 to probe hard samples near the decision boundary and use the L_1 distance as a stable discrepancy measure between them. This strategy has proven effective in prior studies [51], [52]. Meanwhile, a target-domain feature extractor F_t and a target-specific classifier G_t are introduced for the discrimination constraint. Unlike existing approaches, we use F_t as a target-domain structure explorer that mines discriminative structures in an independent feature space, rather than learning domain-invariant representations. In addition, G_t generates stable and reliable pseudolabels. The classifiers G_1, G_2, and G_t are pretrained on the source domain using a standard classification loss. To explore as many target-domain samples as possible, G_1 and G_2 are initialized differently. First, we maximize the discrepancy between G_1 and G_2 on the target domain to identify target samples outside the source decision boundary. Then, following an adversarial learning strategy, we minimize this discrepancy to encourage F_t to learn a discriminative feature distribution aligned with the source domain. Finally, the decision boundary of the main classifier G_t is constrained to lie between those of G_1 and G_2, thereby improving its tolerance to target-domain support vectors. On the one hand, adversarial optimization effectively smooths the pseudolabels; on the other hand, the convergence of G_t between G_1 and G_2 further mitigates noise. As a result, DCM is structurally robust to prediction uncertainty in the early training stage. The discrimination constraint loss $\mathcal{L}_{dis}$ is defined as follows:

$$\begin{aligned} \mathcal{L}_{dis} = \frac{1}{n_t} \sum_{j=1}^{n_t} &\left|\left|G_1\left(F_t\left(x_j^t\right)\right) - G_2\left(F_t\left(x_j^t\right)\right)\right|\right|_1 \\ &+ \left|\left|G_t\left(F_t\left(x_j^t\right)\right) - G_1\left(F_t\left(x_j^t\right)\right)\right|\right|_1 \\ &+ \left|\left|G_t\left(F_t\left(x_j^t\right)\right) - G_2\left(F_t\left(x_j^t\right)\right)\right|\right|_1 \end{aligned} \tag{6}$$

where $\|\cdot\|_1$ denotes L_1 regularization, used to prevent overfitting. It is important to clarify that DCM does not explicitly identify or filter outlier classes. This is because DCM operates only on target-domain features, and under the PDA setting, the target domain inherently contains only classes shared with the source domain, without source-private classes. Therefore, DCM training introduces no interference from outlier classes.

E. Supervised Contrastive Learning and Reweighting

To balance feature transferability and discriminability, we use supervised contrastive learning to coordinate with the domain adversarial network during alignment. Through supervised contrastive learning, the source-biased decision boundary adapts to target data, implicitly guiding adversarial training toward a better saddle point and explicitly calibrating the classification boundary.

First, we use the target-specific classifier G_t to provide reliable supervision signals for target-domain contrastive learning, enabling the model to perceive class information of target samples. By pulling the same class samples closer and pushing different-class samples apart, contrastive learning enables target pseudolabels to guide supervised contrastive learning in explicitly optimizing classification boundaries toward the true distribution, thereby achieving explicit calibration. In particular, for a minibatch of target samples, we define the prediction set as $\mathbf{P}^t = \{p_1^t, \ldots, p_N^t\}$ and construct the pseudolabel set U for the target samples as follows:

$$U = \left\{\left(x_i^t, \hat{y}_i^t = \arg\max\left(p_i^t\right)\right)\right\}, \quad i = 1, 2, \ldots, N. \tag{7}$$

Unlike conventional self-supervised contrastive learning, supervised contrastive learning defines samples sharing the anchor label as positive samples, while those with different labels and their augmentations are treated as negative samples. In particular, we denote the original samples from the source and target domains as $\mathbf{X}^s = \{x_1^s, x_2^s, \ldots, x_N^s\}$ and

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$\mathbf{X}^t = \{x_1^t, x_2^t, \ldots, x_{|U|}^t\}$, respectively, where $|U|$ denotes the cardinality of set U. Each sample is then cropped and recolored to generate augmented samples $\mathbf{X}^{s'} = \{x_1^{s'}, x_2^{s'}, \ldots, x_N^{s'}\}$ and $\mathbf{X}^{t'} = \{x_1^{t'}, x_2^{t'}, \ldots, x_{|U|}^{t'}\}$.

Next, to exploit label information, we adopt supervised contrastive learning to enhance interclass discriminability. Following prior studies [43], [46], the low-dimensional embeddings obtained via linear projection are used as inputs to the contrastive objective. In particular, $\mathbf{P}^{s'} = \{p_1^{s'}, p_2^{s'}, \ldots, p_N^{s'}\}$ and $\mathbf{P}^{s'} = \{p_1^{s'}, p_2^{s'}, \ldots, p_N^{s'}\}$ denote the embedding vectors of source samples and their augmented counterparts after linear mapping, respectively. Similarly, $\mathbf{P}^t = \{p_1^t, p_2^t, \ldots, p_{|U|}^t\}$ and $\mathbf{P}^{t'} = \{p_1^{t'}, p_2^{t'}, \ldots, p_{|U|}^{t'}\}$ represent the embeddings of target samples and their augmented versions. For simplicity, we define $\mathbf{P} = \mathbf{P}^s \cup \mathbf{P}^t$, $\mathbf{P}' = \mathbf{P} \cup \mathbf{P}^{s'} \cup \mathbf{P}^{t'}$. Benefiting from its positive and negative sample construction mechanism, supervised contrastive learning can implicitly capture intraclass and interclass sample relationships across domains, thereby enhancing intraclass similarity and interclass separability. As a result, it provides class-consistency information to the adversarial network, enabling fine-grained class-level alignment to complement coarse-grained domain-level alignment, thus implicitly guiding adversarial training toward a more balanced and meaningful equilibrium. It is worth noting that, since supervised contrastive learning focuses on class-level cross-domain alignment, it does not require explicit filtering strategies for source-domain outlier classes. The supervised contrastive loss is defined as follows:

$$\mathcal{L}_{con} = \sum_{i=1}^{2(N+|U|)} \frac{1}{2|P(i)|-1} \sum_{j \in A(i)} \log \frac{\exp\left(p_i \cdot p'_j/\tau\right)}{\sum_{k \in A(i)} \exp\left(p_i \cdot p'_k/\tau\right)} \tag{8}$$

where $P(i)$ denotes the set of positive samples sharing the same label as the anchor x_i and $|P(i)|$ represents its cardinality. $A(i)$ denotes the negative sample set. τ is the temperature parameter in contrastive learning, used to scale similarity scores.

Finally, to leverage classifiers G and G_t in generating sample weights w, we adopt the dual-level weighting strategy proposed in [13], where the source sample weight w comprises class-level weight w^s and instance-level weight w^t. The class-level weight w^s is derived from predictions of G on target-domain samples. Since the target domain contains no source-private classes, predictions for outlier classes in the output of G are sufficiently low, yielding smaller class-level weights w^s for outlier classes than shared classes. In particular, we first compute the average class prediction w^s over a mini-batch of target-domain samples and then normalize it along the class dimension C_s to obtain the predicted probability of each class, serving as the class-level weight w_i^s. The computation is defined as follows:

$$w^s = \frac{1}{n_t} \sum_{i=1}^{n_t} G\left(F\left(x_i^t\right)\right), i = 1, 2, \ldots, n_t \tag{9}$$

$$w_i^s = w_i^s / \sum_{i=1}^{C_s} w_i^s, i = 1, 2, \ldots, C_s \tag{10}$$

where C_s denotes the number of classes in the source domain.

The instance-level weights w^t are obtained from the prediction probabilities of G_t on source samples. Since G_t is a target-specific classifier and has not seen source-private class information, it assigns significantly lower weights to source outlier samples than shared samples. In particular, we first compute the softmax-activated prediction vector $\hat{y}_i^s$ for each source instance and then calculate the instance-level weight w_i^t using information entropy. The computation is given as follows:

$$\hat{y}_i^s = G_t\left(F_t\left(x_i^s\right)\right), i = 1, 2, \ldots, n_s \tag{11}$$

$$w_i^t = 1 + e^{-H\left(\hat{y}_i^s\right)}, i = 1, 2, \ldots, n_s \tag{12}$$

where $-H(\cdot)$ denotes the entropy-based criterion used to select more reliable instances.

By combining these two weight levels and assigning them to the domain adversarial network, the model becomes significantly less sensitive to source-private class samples during training, thereby mitigating negative transfer. It is worth noting that the main classifier G produces class-level weights, while the target-specific classifier G_t in DCM generates instance-level weights. Fusing these two weight types suppresses the interference of outlier classes and highly noisy samples during early training, thereby improving adversarial alignment stability. The formulation for weight fusion is given as follows:

$$w_i = w_{d(i)}^s w_i^t, \quad i = 1, 2, \ldots, n_s \tag{13}$$

where $d(i)$ denotes the class index of the ith sample and n_s is the total number of source samples. Since the normalized class-level weight w_i^s is constrained to $[0, 1]$, and the instance-level weight w_i^t is typically close to 1, the combined weight w_i naturally falls within $[0, 1]$.

F. Overall Objective and Training

Combining (2), (3), (6), and (8), the overall loss of the proposed method, $\mathcal{L}_{overall}$, is summarized as follows:

$$\mathcal{L}_{overall} = \mathcal{L}_{adv} + \mathcal{L}_{pow} + \beta \mathcal{L}_{dis} + \mathcal{L}_{con} \tag{14}$$

where β is a hyperparameter balancing the contribution of the DCM. The training process of our method is formulated as a min–max optimization problem

$$\begin{aligned} \left(\hat{\theta}_f, \hat{\theta}_g, \hat{\theta}_{f_t}, \hat{\theta}_{g_t}\right) &= \arg\min_{\theta_f, \theta_g, \theta_{f_t}, \theta_{g_t}} \left(\theta_f, \theta_g, \hat{\theta}_d, \theta_{f_t}, \theta_{g_t}, \hat{\theta}_{g_1}, \hat{\theta}_{g_2}\right) \\ \left(\hat{\theta}_d, \hat{\theta}_{g_1}, \hat{\theta}_{g_2}\right) &= \arg\max_{\theta_d, \theta_{g_1}, \theta_{g_2}} \left(\hat{\theta}_f, \hat{\theta}_g, \theta_d, \hat{\theta}_{f_t}, \hat{\theta}_{g_t}, \theta_{g_1}, \theta_{g_2}\right) \end{aligned} \tag{15}$$

where θ_f, θ_g, θ_d, θ_{f_t}, θ_{g_t}, θ_{g_1}, and θ_{g_2} denote the parameters of the feature extractor F, classifier G, domain discriminator D, auxiliary feature extractor F_t, target-domain-specific classifier G_t, and auxiliary classifiers G_1 and G_2. These parameters are optimized through backpropagation. The auxiliary feature extractor F_t does not share parameters with the shared feature extractor F, and their gradient update paths remain independent during optimization. This decoupled optimization strategy avoids gradient conflicts between different objectives, thereby ensuring training stability. For clarity, we summarize the training procedure of SCAPDA and the training parameters in Algorithm 1.

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Algorithm 1 SCAPDA

Input: The source domain D_s and the target domain D_t; The feature extractor F; The domain discriminator D; The classifier G; The auxiliary classifiers G_1 and G_2; The target-specific classifier G_t; The validation T.

Output: The trained feature extractor F; The trained classifier G.

Pre-train the feature extractor F on ImageNet dataset; Pre-train the auxiliary classifiers G_1 and G_2 on labeled source samples differently;
while not converge **do**
  Initialize the class-level weight w^s;
  for $i = 1 : T$ **do**
    Sample a source domain minibatch $\{(x_i^s, y_i^s)\}_{i=1}^N$ from D_s;
    Obtain the augmentation of $\{(x_i^s, y_i^s)\}_{i=1}^N$
    Sample a target domain minibatch $\{(x_i^t)\}_{i=1}^N$ from D_t
    Obtain the augmentation of $\{x_i^t\}_{i=1}^N$;
    Compute the instance-level weight w^s by the output of G_t;
    Compute the final weight w by Eq. 13;
    Compute the adversarial network loss by Eq. 2 and Eq. 3;
    Construct the pseudo-label set U for target domain by Eq. 7;
    Compute the discrimination constraint loss by Eq. 6;
    Compute the supervised contrastive learning loss bt Eq. 8;
    Update parameters θ_f, θ_g, θ_d, θ_{f_t}, θ_{g_t}, θ_{g_1} and θ_{g_2};
  end for
end while

IV. Experiments and Analyses

A. Data Description

We use three publicly available RSSC datasets to construct PDA tasks and evaluate the proposed method: the AID dataset [53], the UC Merced dataset [54], and the NWPU-RESISC45 dataset [55].

The AID dataset consists of 30 aerial scene classes collected from the Google Earth platform, comprising 10 000 images captured across various countries, climates, and imaging conditions. It serves as a standardized benchmark for evaluating RSSC methods. Each image has a resolution of 600×600 pixels, with spatial resolutions ranging from 8 to 0.5 m.

The UC Merced dataset consists of 21 land-use classes extracted from aerial imagery provided by the U.S. Geological Survey, totaling 2100 images. Each class contains 100 images, all with a fixed resolution of 256×6256 pixels and a spatial resolution of 1 ft.

The NWPU-RESISC45 dataset consists of 45 scene classes collected from the Google Earth platform, with 31 500 images in total. Each class contains 700 images, and all images have a fixed size of 256×6256 pixels. The spatial resolution ranges from 30 to 0.2 m.

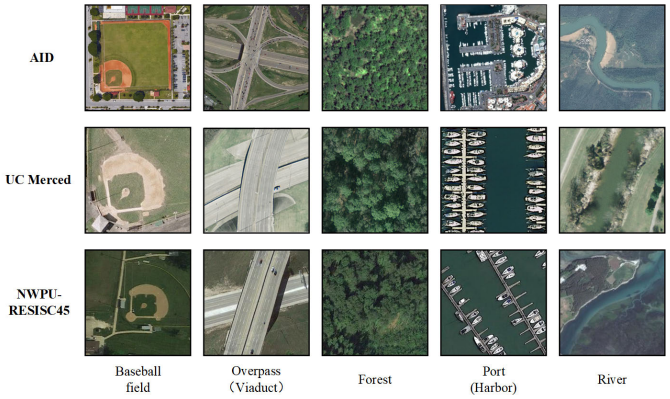


Fig. 4. Example classes of dataset, including AID, UC Merced, and NWPU-RESISC45.

Based on these datasets, we construct six RSSC tasks under the PDA setting. The shared and source-private classes for each task are summarized in Table II, where semantically identical but differently named scene classes are indicated in parentheses. In general, tasks with more source-private classes are more difficult to align with the target domain. It is worth noting that in the task transferring from UC Merced to NWPU-RESISC45, all UC Merced classes are included in NWPU-RESISC45. Therefore, this task becomes a UDA problem. Although such a closed-set scenario is less common, it may occur in real-world applications. We retain this task to comprehensively evaluate the generalization performance of our method. Fig. 4 presents example images from the six tasks, showing substantial distribution differences caused by variations in sensors, resolutions, and spectral ranges, as well as high interclass similarity among scene categories. These challenges make the constructed scenarios suitable for evaluating the ability of our method to balance discriminability and transferability in PDA settings.

B. Implementation Details

We follow the standard unsupervised PDA protocol, using fully labeled source-domain data and unlabeled target-domain data. The average classification accuracy (ACC) is adopted as the evaluation metric. In our experiments, ResNet-50 [56] is employed as the baseline architecture. The proposed SCAPDA and other compared methods are implemented with PyTorch on a Tesla V100 GPU. We resize input images to 224×6224 pixels. Similar to [19], the ResNet-50 backbone is pretrained on ImageNet [57]. We replace the last fully connected layer of the pretrained model with a 256-D bottleneck layer as the feature extractor F. The adversarial discriminator penalty is gradually increased from 0 to 1 to ensure training stability. The proposed model is optimized via backpropagation using stochastic gradient descent (SGD) [58] with a momentum of 0.9. The learning rate is set to 0.001, with a scheduling strategy similar to that in [9]. In addition, we set the batch size of each method to 36. The parameter α of the uncertainty function $\mathcal{H}_\alpha(\mathbf{p})$ is set to 6. For SCLM, we adopt a dual-view data augmentation strategy, where one view is generated using standard data transformations, and the other augmented view includes random cropping and color perturbation. We conduct

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TABLE II
CLASSES INFORMATION FOR SIX PDA TASKS

Source Domain	Target Domain	Share Classes	Source Private Classes
AID	UC Merced	Baseball field (Baseball diamond), Beach, Commercial (Buildings), Dense residential, Forest, Medium residential, Parking, Port (Harbor), Sparse residential, Storage tanks, Viaduct (Overpass), Farmland (Agricultural), River	Airport, Bare land, Bridge, Square, Stadium, Center, Church, Desert, School, Mountain, Industrial, Meadow, Railway station, Park, Playground, Pond, Resort
AID	NWPU-RESISC45	Airport, Baseball field, Beach, Bridge, Church, Forest, Commercial, Dense residential, Medium residential, Desert, Farmland (Circular farmland & Rectangular farmland), Playground (Ground track field), Port (Harbor), Industrial, Sparse residential, Storage tanks, Viaduct (Overpass), Meadow, Mountain, River, Railway station, Stadium, Parking	Bare land, Center, School, Park, Pond, Resort, Square
UC Merced	AID	Agricultural (Farmland), Baseball diamond (Baseball field), Buildings (Commercial),Dense residential, Forest, Harbor (Port), Medium residential, Beach,Overpass (Viaduct), Parking lot (Parking), River, Sparse residential, Storage tanks	Airplane, Chaparral, Golf course, Mobile home park, Tennis court, Intersection, Runway, Freeway
UC Merced	NWPU-RESISC45	Agricultural (Circular farmland & Rectangle farmland), Airplane, Baseball diamond,Beach, Buildings (Commercial), Dense residential, Forest, Freeway, Golf course,Harbor, Intersection, Medium residential, Mobile home park,Overpass, Parking lot, River,Runway, Tennis court, Storage tanks, Sparse residential, Chaparral	None
NWPU-RESISC45	AID	Airport, Baseball diamond (Baseball field), Beach, Church, Commercial area, Circular farmland & Rectangle farmland (Farmland), Harbor (Port), Railway station, River, Sparse residential, Dense residential, Parking lot, Ground track field (Playground), Forest, Industrial area, Meadow, Medium residential, Mountain, Stadium, Storage tank, Overpass (Viaduct), Desert, Bridge	Basketball court, Chaparral, Cloud, Thermal power station, Freeway, Golf course, Intersection, Island, Terrace, Airplane, Lake, Mobile home park, Palace, Tennis court, Roundabout, Runway, Sea ice, Ship, Snowberg, Wetland
NWPU-RESISC45	UC Merced	Airplane, Baseball diamond, Beach, Chaparral, Commercial area, Dense residential, Forest, Freeway, Golf course, Harbor, Intersection, Circular farmland & Rectangular farmland (Farmland), Mobile home park, Parking lot, River, Runway, Sparse residential, Storage tank, Tennis court, Medium residential, Overpass	Airport, Basketball court, Bridge, Church, Island, Lake, Cloud, Desert, Ground track field, Industrial area, Meadow, Mountain, Palace, Railway, Railway station, Sea ice, Ship, Snowberg, Stadium, Terrace, Wetland, Roundabout, Thermal power station

TABLE III
ACCURACY (%) ON OUR COLLECTED DATASET FOR PDA SCENARIOS

	Method	A→U	A→N	U→A	U→N	N→A	N→U	Avg
	ResNet-50 [56]	51.38±0.96	72.32±0.74	57.74±0.85	60.21±1.12	80.43±0.94	73.67±0.59	65.96±0.87
UDA	DANN [9]	50.46	75.68	65.58	64.76	77.04	63.00	66.09
	CDAN [59]	46.62	74.62	64.66	64.81	76.46	62.71	64.98
	TransResNet [60]	51.15	75.38	64.64	63.93	81.11	73.29	68.25
	Caco [61]	56.46	73.45	55.67	62.26	82.75	72.81	67.23
	SFMDA [62]	60.15±0.23	78.38±0.18	65.94±0.15	73.05±0.20	84.29±0.09	75.44±0.25	72.88±0.18
UniDA	SDG-MA [63]	45.31	64.97	54.64	56.03	64.92	52.72	54.23
	C^3DA [64]	51.57	60.13	55.97	70.71	65.74	55.33	58.81
PDA	SAN [11]	58.08	79.55	76.46	68.81	85.93	76.43	74.21
	IWAN [17]	60.00	75.08	66.50	67.42	81.58	79.29	71.65
	PADA [18]	51.85	72.02	61.08	59.23	83.14	74.14	66.91
	ETN [19]	52.08	78.67	71.45	68.50	84.90	74.76	71.73
	CPADA [6]	59.58	72.92	73.47	67.01	85.49	73.88	72.06
	BA3US [65]	58.77±0.11	83.98±0.06	73.74±0.16	73.70±0.37	89.21±0.31	75.86±0.29	75.88±0.22
	DRCN [20]	64.15	79.18	63.89	66.71	85.81	80.67	73.40
	CLA [13]	63.62±0.13	84.17±0.11	76.31±0.28	74.19±0.29	85.66±0.10	80.82±0.12	77.46±0.17
	I-DANN [7]	64.62	84.61	**79.71**	**76.92**	89.54	80.76	79.36
	SLM [14]	62.74±0.25	81.41±0.21	75.23±0.33	74.06±0.19	86.38±0.14	79.73±0.54	76.93±0.28
	STCPDA [23]	63.24±0.19	80.58±0.07	76.70±0.28	72.55±0.32	85.21±0.23	80.15±0.40	77.32±0.25
	AR [24]	63.23±0.08	84.67±0.10	77.11±0.17	71.80±0.21	85.66±0.09	79.85±0.28	77.14±0.16
	Ours	**65.86**±0.07	**88.54**±0.12	79.62±0.10	76.57±0.11	**90.77**±0.06	**82.25**±0.14	**80.60**±0.10

experiments on six transfer tasks and report the ACC. The six tasks are A→U, A→N, U→A, U→N, N→A, and N→U, where A, U, and N denote the AID, UC Merced, and NWPU-RESISC45 datasets, respectively. Note that U→N corresponds to the standard closed-set UDA setting.

C. Comparison With State-of-the-Art Methods

To evaluate the effectiveness of SCAPDA, we compare it with 12 state-of-the-art PDA approaches, including SAN [11], IWAN [17], PADA [18], ETN [19], CPADA [6], BA3US [65], DRCN [20], CLA [13], I-DANN [7], SLM [14], STCPDA [23], and AR [24]. To further demonstrate the negative impact of outlier classes, we also compare our method with five standard domain adaptation (UDA) methods under the PDA setting, including DANN [9], CDAN [59], TransResNet [60], Caco [61], and SFMDA [62]. ResNet-50, trained only on the source domain, is included as a baseline to further validate the effectiveness of domain adaptation methods. In addition, to demonstrate the superiority of our method in handling

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TABLE IV
ACCURACY (%) OF ABLATION STUDIES ON OUR COLLECTED DATASET

Method	A→U	A→N	U→A	U→N	N→A	N→U	Avg
SCAPDA/SCLM	60.31±0.19	77.82±0.12	75.47±0.09	70.16±0.22	86.96±0.17	77.78±0.23	74.75±0.17
SCAPDA/DCM	63.95±0.11	82.03±0.06	78.62±0.08	73.74±0.15	88.72±0.03	79.98±0.18	77.84±0.10
SCAPDA/w^t	65.20±0.05	84.25±0.08	78.58±0.10	75.13±0.12	89.41±0.09	80.77±0.08	79.22±0.09
SCAPDA/$\mathcal{H}_\alpha$	62.43±0.08	82.29±0.15	77.23±0.14	74.52±0.05	88.45±0.13	78.21±0.25	77.19±0.13
SCAPDA	**65.86**±0.07	**88.54**±0.12	**79.62**±0.10	**76.57**±0.11	**90.77**±0.06	**82.25**±0.14	**80.60**±0.10

PDA tasks over existing universal domain adaptation (UniDA) methods, we compare our method with SDG-MA [63] and C^{3}DA [64], which are applied to RSSC. The classification results across the six PDA tasks are presented in Table III. To reduce the influence of randomness, we conduct five independent runs for our method and several comparison methods. The best results are highlighted in bold, and the second-best results are underlined.

According to the results, SCAPDA achieves the best performance on most tasks, with an average accuracy of 80.60%. It outperforms current SOTA PDA methods by 1.24%–13.69% and achieves an average improvement of 1.97% over the RSSC PDA method I-DANN across four tasks. Although slight performance drops of 0.09% and 0.35% are observed on U→A and U→N, respectively, this can be attributed to I-DANN performing progressive alignment by constructing auxiliary domains from the source domain and relying more on stable source-domain features in early training, thereby showing advantages on certain tasks. On the contrary, SCAPDA emphasizes the balance between discriminability and transferability, together with discriminative constraints on the target domain, which may cause minor fluctuations under extreme domain shifts. Nevertheless, its overall superiority across all tasks demonstrates that incorporating discriminative information can substantially enhance cross-domain transfer capability.

In comparison, UDA methods perform poorly under the PDA setting. For instance, in the N→U task, except for the latest SFMDA method, all other UDA approaches cause an average accuracy degradation of 0.38%–10.96% compared with ResNet-50. On average, PDA methods achieve higher classification performance than standard UDA methods, confirming the effectiveness of filtering source-private classes to mitigate negative transfer. On the UDA task U→N, our method still achieves a significant advantage, outperforming models specifically designed for UDA by 7.72%–15.62%, demonstrating strong generalization capability in conventional UDA scenarios. Moreover, compared with general DA methods for RSSC, namely SDG-MA and C^{3}DA, our method improves average accuracy by 21.79%–26.37%, showing that UniDA methods struggle with class-space mismatch and negative transfer in PDA settings and highlighting the need for PDA-specific models.

In summary, the results indicate that UDA methods suffer noticeable performance degradation on PDA tasks. This is mainly due to introducing source-domain outlier classes into adversarial training, which causes negative transfer. Recent PDA methods address this issue by detecting and filtering source-only samples, significantly improving classification performance on tasks such as A→N and U→A. These results show that removing outlier classes effectively mitigates negative transfer. Our method considers high interclass similarity in complex remote sensing scenes, further explores the discriminative structure of the target domain, and leverages this information to balance transferability and discriminability. As a result, our method achieves the best performance on most PDA tasks. These findings confirm that SCAPDA enables effective knowledge transfer in complex PDA scenarios and enhances the model's discriminability for challenging classes across domains.

D. Discussions and Analyses

1) Ablation Studies: To further evaluate each component of the proposed method, we compare it with four variants. In particular, SCAPDA/SCLM denotes a variant without supervised contrastive learning coordination; SCAPDA/DCM replaces the DCM with the target clustering strategy from CLA; SCAPDA/w^t removes instance-level weighting in the reweighting strategy; and SCAPDA/$\mathcal{H}_\alpha$ is a variant without prediction uncertainty modeling. Table IV presents the classification accuracies of SCAPDA and its variants on six PDA tasks. Each setting is evaluated over five independent runs to reduce randomness.

The results show that our method significantly outperforms SCAPDA/SCLM, improving average accuracy by 5.85%. This indicates that supervised contrastive learning effectively guides adversarial training toward a better convergence point, enhancing model discriminability while promoting transferability. Consequently, the model makes more accurate classification decisions on complex remote sensing images. In addition, our method outperforms SCAPDA/DCM, verifying that DCM better captures the intrinsic structure of the target domain and produces higher quality pseudolabels than the target clustering strategy in CLA. Furthermore, our method exceeds SCAPDA/w^t, suggesting that instance-level weights for target samples enable more accurate identification and suppression of source-domain outlier classes. Compared with SCAPDA/$\mathcal{H}_\alpha$, our approach still achieves superior results, validating the integration of the uncertainty-aware weighting function. This module reduces the expected error on target samples by embedding prediction uncertainty into the training framework.

We further replace the adversarial network with metric-matching alignment methods to evaluate adversarial training in complex remote sensing scenarios. These metric-based approaches include maximum mean discrepancy (MMD), correlation alignment (CORAL), Kullback–Leibler (KL)

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TABLE V
ACCURACY (%) OF ADVERSARIAL NETWORKS VERSUS METRIC-BASED ALIGNMENT METHODS

Method	A→U	A→N	U→A	U→N	Avg
MMD	63.73	82.04	75.42	74.21	73.85
CORAL	61.27	81.39	77.12	73.73	73.38
KL Divergence	63.51	82.84	77.26	75.95	74.74
Wasserstein Distance	64.10	83.45	76.73	76.35	75.12
SCAPDA	**65.86**	**88.54**	**79.62**	**76.57**	**77.65**

divergence, and Wasserstein distance. Table V summarizes the results. Although metric-based methods offer better interpretability and computational efficiency, they may be less suited for complex real-world remote sensing scenarios. As shown, our adversarial alignment strategy significantly outperforms the other approaches, improving average accuracy by 2.53%–4.27%. This improvement stems from the superior capability of the adversarial network in capturing and matching complex, high-dimensional data distributions. On the contrary, metric-based methods rely on linear modeling and specific distributional assumptions, hindering reliable domain alignment under complex scene conditions.

To further investigate synergy among different modules, we design various module combinations and conduct ablation studies, with results presented in Table VI. Each setting is evaluated over five independent runs to reduce randomness. The baseline network adopts domain adversarial training with only class-level reweighting. Benefiting from SCLM synergy, the average accuracy improves by 4.85% over the plain adversarial network, indicating enhanced transferability and discriminability. Moreover, SCLM shows strong synergy with DCM, where DCM provides high-confidence supervisory signals and instance-level weights to further suppress negative transfer. Finally, incorporating the uncertainty function $\mathcal{H}_\alpha$ yields an additional 3.88% average accuracy improvement by enhancing interclass separation, intraclass compactness, and classifier robustness on boundary samples.

To verify the effectiveness of the double-classifier structure in DCM, we conduct experiments by removing or adding auxiliary classifiers that constrain the target-domain discriminative structure through classifier discrepancy. The results are presented in Table VII. In particular, DCM/single, DCM/double, and DCM/triple denote variants with one, two, and three auxiliary classifiers, respectively. According to Table VII, the double-classifier design achieves the best performance, outperforming the other two variants by 2.49%–11.13% in average accuracy. This advantage can be attributed to the ability of two classifiers to efficiently generate discrepancy signals relative to the source domain, thereby maximizing the detection of target samples outside the source distribution. On the contrary, the single-classifier variant fails to produce discrepancy signals, making DCM ineffective and unable to provide valid supervision for SCLM. Meanwhile, the triple-classifier setting may cause a voting effect, which can obscure genuine discrepancies in the target domain and hinder the detection of domain shifts beyond the source distribution.

In addition, we conduct ablation experiments to evaluate the early-stage stability of DCM and systematically analyze its

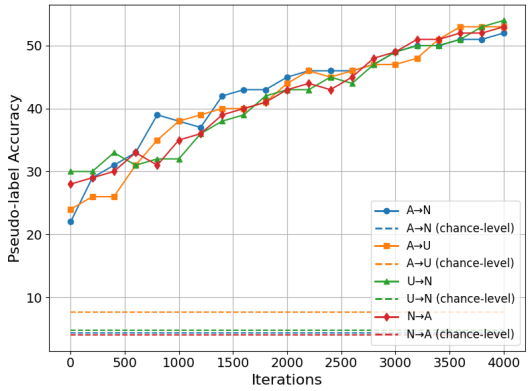


Fig. 5. Pseudolabel accuracy of DCM over early-stage (4000 iterations).

sensitivity to pseudolabel noise. In particular, for four transfer tasks, we record pseudolabel accuracy during the first 4000 iterations. As illustrated in Fig. 5, the pseudolabel accuracy for all tasks shows a stable upward trend from the initial stage with only minor fluctuations and remains significantly higher than random guessing, indicated by the dashed lines (14.30%–49.24%). According to empirical evidence reported in related studies [66], [67], these observations indicate that DCM can provide sufficiently reliable supervisory signals even at an early training stage, thereby validating the effectiveness of the max–min discrepancy mechanism and double-classifier consistency constraint in suppressing noisy pseudolabels.

2) Computational Complexity: To investigate the tradeoff between computational complexity and model performance of SCAPDA, we use params and MACs as complexity metrics and compare our method with several SOTA approaches. The quantitative results are summarized in Table VIII. As shown, our method has fewer parameters than SLM and CLA while achieving higher ACC across the four PDA tasks by 4.29% and 3.07%. These results demonstrate the superiority of our approach in both complexity and performance. The performance gains of SCAPDA arise mainly from two aspects: 1) DCM provides more reliable pseudolabels for target-domain samples, effectively alleviating error accumulation; 2) the collaborative optimization between SCLM and adversarial learning further enhances discriminative feature transfer quality. Although SCAPDA introduces slightly more parameters than STCPDA, it improves the average accuracy by 4.38%. This is because STCPDA adopts an MMD-based alignment strategy with simple pseudolabel selection, which easily introduces noise and error accumulation during self-training. On the contrary, our DCM leverages a double-classifier mechanism to explore target-domain discriminative structures and provide high-confidence supervisory signals for SCLM, effectively suppressing pseudolabel noise with limited computational overhead.

It is worth noting that the overall parameter count of our method is only marginally higher than that of STCPDA while remaining lower than those of SLM and CLA. Therefore, SCAPDA can be trained and inferred within a reasonable time on modern GPU platforms. Moreover, during deployment, only the target-specific classifier G_t is required for inference, without continuously running the auxiliary classifiers used

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TABLE VI
ACCURACY (%) OF THE SYNERGY BY COMBINING DIFFERENT KEY COMPONENTS

Backbone	Components				A→U	A→N	U→A	U→N	Avg
	SCLM	DCM	w^t	$\mathcal{H}_\alpha$					
ResNet50	✗	✗	✗	✗	57.08±0.16	75.93±0.11	72.19±0.12	65.61±0.07	67.70±0.12
	✓	✗	✗	✗	61.52±0.08	80.63±0.14	76.27±0.13	71.78±0.15	72.55±0.13
	✓	✗	✗	✓	62.16±0.12	81.77±0.15	78.32±0.06	72.85±0.09	73.77±0.11
	✓	✓	✗	✗	62.04±0.11	80.92±0.08	76.84±0.15	73.79±0.06	73.40±0.10
	✓	✓	✓	✗	62.43±0.09	82.29±0.06	77.23±0.17	74.52±0.14	74.12±0.12
	✓	✓	✓	✓	**65.86**±0.07	**88.54**±0.12	**79.62**±0.10	**76.57**±0.11	**77.65**±0.10

TABLE VII
ACCURACY (%) OF DCM WITH DIFFERENT NUMBERS OF AUXILIARY CLASSIFIERS

Method	A→U	A→N	U→A	U→N	Avg
DCM/single	55.19	76.84	70.47	63.58	66.52
DCM/double	**65.86**	**88.54**	**79.62**	**76.57**	**77.65**
DCM/triple	62.78	84.84	79.51	73.52	75.16

TABLE VIII
COMPARISON OF PARAMETER NUMBER, COMPUTATIONAL COMPLEXITY, AND AVERAGE ACCURACY (%)

Method	A→U	A→N	U→A	U→N	Avg	Params	MACs
ResNet-50 [56]	51.38	72.32	57.74	60.21	60.41	23.57M	8.26G
CLA [13]	63.62	84.17	76.31	74.19	74.58	32.16M	9.05G
STCPDA [23]	63.24	80.58	76.70	72.55	73.27	28.30M	8.68G
SLM [14]	62.74	81.41	75.23	74.06	73.36	32.29M	9.25G
Ours	**65.86**	**88.54**	**79.62**	**76.57**	**77.65**	32.14M	9.37G

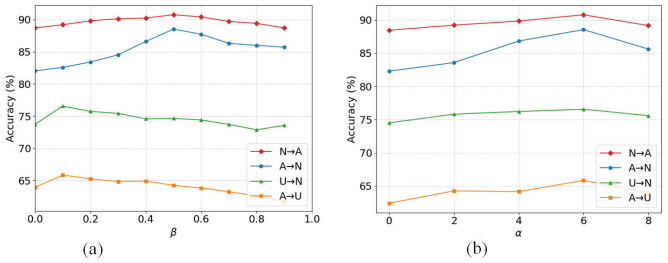


Fig. 6. Sensitivity analysis for the four tasks with respect to (a) hyperparameter β and (b) parameter α.

in the max–min discrepancy scheme, which further reduces computational cost. Consequently, although DCM introduces additional classifiers, it achieves an effective tradeoff between performance and computational complexity and offers a novel perspective for mitigating pseudolabel noise in PDA.

3) Parameter Analysis: To investigate contrastive learning coordination within the adversarial network, we conduct a sensitivity analysis of the hyperparameter β on four transfer tasks: A→U, A→N, U→N, and N→A. We vary β over [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9] and observe the corresponding performance changes. The results are presented in Fig. 6(a).

The results show that the hyperparameter β exhibits different coordination abilities across tasks. For example, the A→U and U→N tasks achieve optimal performance when $\beta \approx 0.1$, whereas A→N and N→A perform best when $\beta \approx 0.5$. This variation arises from changes in domain distribution

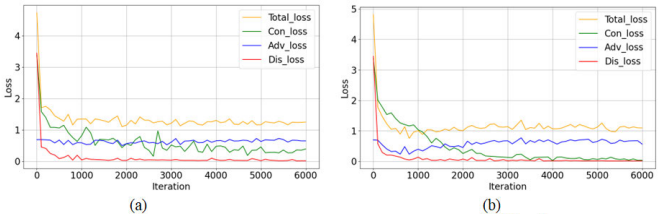


Fig. 7. Loss curve on two tasks. (a) A→N task. (b) A→U task. The four losses are the total loss, contrastive loss, adversarial loss, and discrimination constraint loss.

discrepancies. Due to inherent distribution differences, different remote sensing scene datasets exhibit varying degrees of interclass similarity and intraclass diversity as the transfer task changes. Consequently, supervised contrastive learning requires different coordination levels, which is reasonable and expected. When β is too low, the contrastive learning component cannot effectively coordinate adversarial training, leading to insufficient model discriminability. Conversely, when β is too high, supervised contrastive learning may dominate training, amplifying pseudolabeling errors and causing misalignment, which degrades transferability. In tasks such as A→U and U→N, where interclass variations are more pronounced, a lower β helps prevent noisy pseudolabel propagation and ensures stable transfer. On the contrary, for tasks such as A→N and N→A, pseudolabels exhibit higher confidence, and class similarity is more evident, allowing a higher β to strengthen discriminative learning. It is worth noting that tuning β is relatively straightforward in practical applications. Based on our training experience, model performance remains stable when $\beta \in [0.1, 0.6]$, requiring only coarse-grained value selection.

We further analyze the effect of the parameter α in the uncertainty function $\mathcal{H}_\alpha(\mathbf{p})$ by setting α to [1, 2, 4, 6, 8]. The case of $\alpha = 0$ is not considered, as it only reflects the number of nonzero probability entries and provides no meaningful information for performance evaluation. The experiments are conducted on four domain adaptation tasks: A→U, A→N, U→N, and N→A, with corresponding results shown in Fig. 6(b). We observe that when $\alpha = 1$, all four tasks achieve the worst performance, since $\mathcal{H}_\alpha(\mathbf{p})$ is ineffective in reducing target-domain prediction uncertainty. When $\alpha = 6$, four tasks generally achieve the best performance, as samples with higher prediction uncertainty have smaller $\mathcal{H}_\alpha(\mathbf{p})$ gradients, implying

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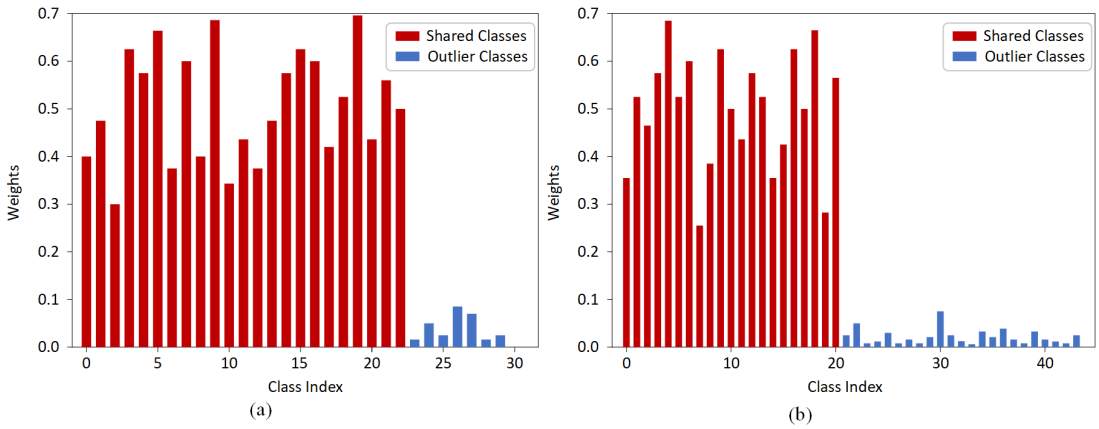


Fig. 8. Illustration of class-level weights for two PDA tasks, where (a) is A→N and (b) is N→U.

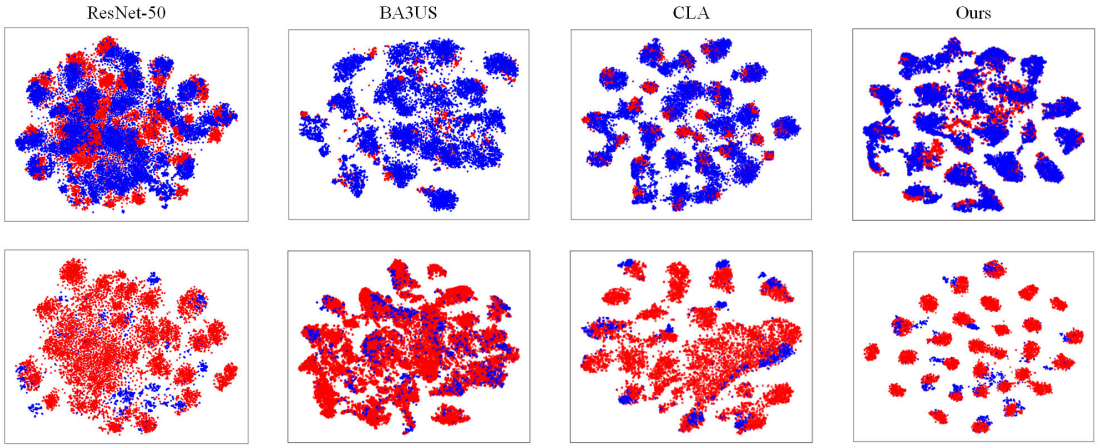


Fig. 9. T-SNE visualization of features for two PDA tasks (top) A→N and (bottom) A→U.

that these samples contribute little to training. Therefore, we set $\alpha = 6$ to minimize target-domain prediction uncertainty.

Based on the above analysis, we determine α and β according to task-specific requirements for transferability and discriminability, as well as overall performance, thereby achieving the optimal balance.

4) Convergence Performance Analysis: To investigate the convergence performance of SCAPDA and synergy among its modules, we conduct convergence experiments on the loss functions for the A→N and A→U tasks, with results shown in Fig. 7. The figure illustrates the training evolution of the loss curves, including the total loss, contrastive loss, adversarial loss, and discrimination constraint loss, as well as their relationships. From Fig. 7(a), all losses tend to converge to a stable state as training iterations increase. In the early training stage, the adversarial loss decreases slowly and remains around 0.4, while the reduction of the total loss mainly stems from the SCLM. In the later stage, both adversarial loss and contrastive loss converge, and the total loss stabilizes around 1.5 under their joint effect. Similarly, Fig. 7(b) shows that in the later training stage, the total loss gradually stabilizes under the synergy of adversarial loss and contrastive loss, further verifying their synergistic effect.

5) Weights Visualization: To evaluate the proposed reweighting mechanism, we analyze the class-level weights assigned in tasks A→N and N→U, as illustrated in Fig. 8. In each subfigure, red bars represent the predicted weights assigned to shared classes between the source and target domains, while blue bars correspond to source-private (outlier) classes. The results show that SCAPDA consistently assigns significantly higher weights to shared classes, whereas the weights of source-private classes remain below 0.1. This clear contrast indicates that the model places minimal emphasis on untransferable outlier classes, thereby constraining adaptation within the shared label space. These findings confirm that our method can accurately identify and suppress source-private classes in PDA tasks.

6) Feature Visualization: In addition, to investigate the effectiveness of our method in aligning source and target domain features and calibrating decision boundaries, we perform t-SNE [68] visualizations on features extracted by the backbone ResNet-50, the baseline methods BA3US and CLA, and the proposed SCAPDA under two PDA tasks. The results are presented in Fig. 9. In each subfigure, high-level semantic features from the source domain are represented by blue points, while those from the target domain are represented by red points.

As shown in Fig. 9, ResNet-50 exhibits scattered feature distributions with severe cross-domain misalignment and unclear decision boundaries. BA3US improves class separation through domain alignment and entropy minimization, but confusion between outlier and shared classes remains.

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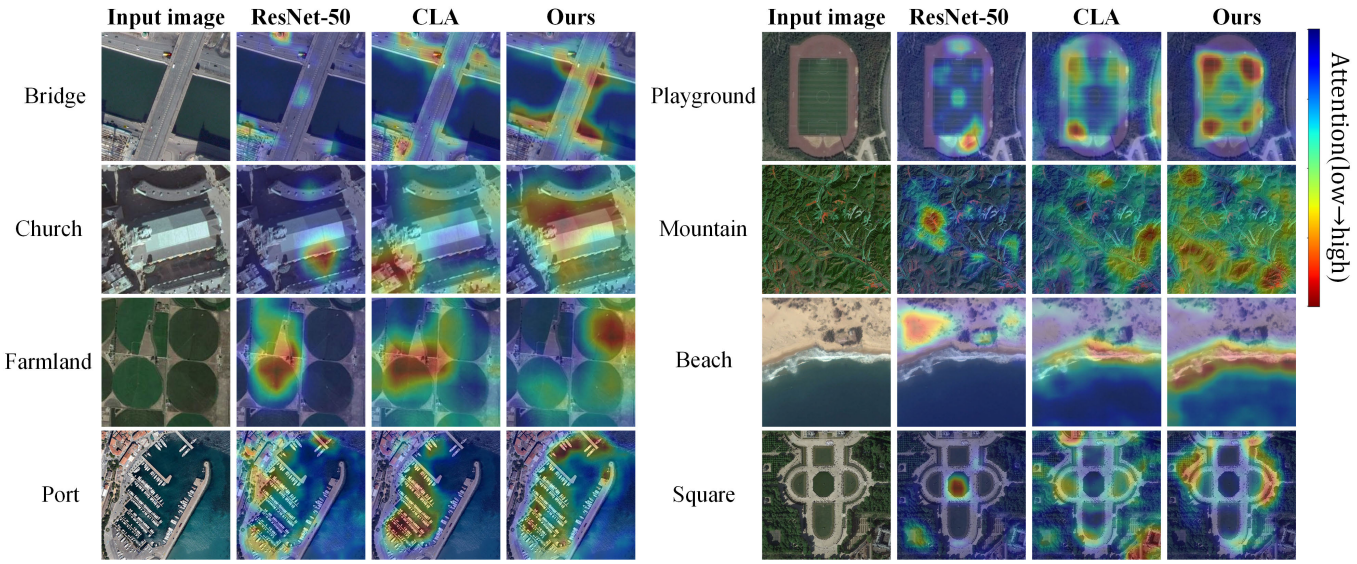


Fig. 10. Grad-CAM++ visualization results. Input images (columns 1 and 5), visualization results from ResNet-50 (columns 2 and 6), visualization results from the CLA (columns 3 and 7), and visualization results from our SCAPDA (columns 4 and 8).

CLA further enhances alignment via dual weighting and class-conditional adversarial alignment, yielding more compact shared-class clusters, although pseudolabel errors still blur decision boundaries. On the contrary, our method exhibits the most distinguishable class structures across both tasks, significantly reducing negative transfer. These results demonstrate that our method enhances feature transferability and calibrates decision boundaries, thereby improving model discriminability.

To further illustrate how our method localizes discriminative features, we employ Grad-CAM++ [69] to visualize attention maps and compare SCAPDA with ResNet-50 and CLA across eight scene classes, as shown in Fig. 10. In particular, in the bridge scene, SCAPDA focuses more on global spectral features, achieving a better balance between transferability and discriminability. In the church scene, it attends more accurately to discriminative building regions while suppressing irrelevant backgrounds. For visually similar scenes such as farmland and mountain, SCAPDA better distinguishes scene-specific texture structures. Overall, these visualizations indicate that the proposed method effectively captures discriminative features across different scenes, achieving a well-balanced tradeoff between transferability and cross-domain class-level discriminability.

7) CM Visualization: To further analyze target-domain classification performance, we present confusion matrices for four PDA tasks, i.e., A→N, A→U, U→A, and U→N. As shown in Fig. 11(a)–(d), our method shows a more concentrated diagonal distribution across all tasks, indicating improved classification accuracy on shared target-domain classes. Meanwhile, the off-diagonal elements are noticeably reduced, suggesting effectively alleviated interclass confusion.

From the perspective of transferability, our method significantly reduces misclassification of target-domain samples into source-private classes. This demonstrates that the model better focuses on shared categories during cross-domain alignment, thereby enhancing feature transferability. In terms of

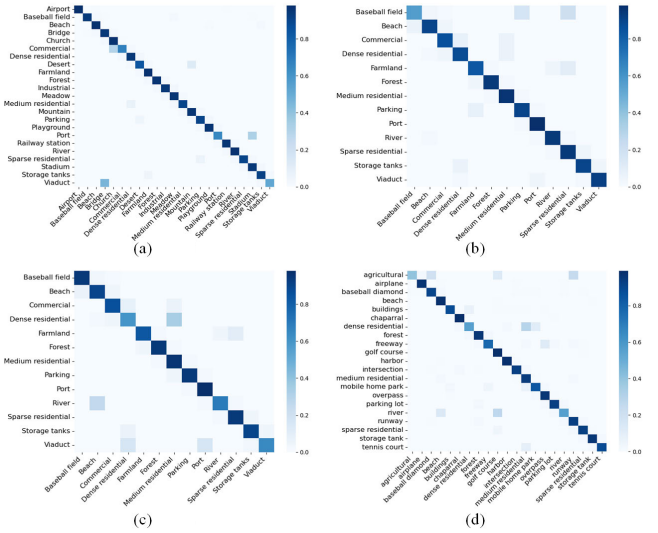


Fig. 11. Confusion matrices for four PDA tasks. (a) A→N. (b) A→U. (c) U→A. (d) U→N.

discriminability, the proposed method achieves clearer distinctions among several visually similar remote sensing scene categories. For example, confusion between visually similar categories, such as river and beach or viaduct and bridge, is substantially reduced, while interclass separability in complex urban scenes is also improved. These improvements mainly stem from the effective balance between transferability and discriminability in our method, enabling the model to enhance class discrimination while maintaining strong cross-domain transfer capability.

8) Consistency in Transferability and Discriminability: To validate the effectiveness of our method in balancing transferability and discriminability in PDA tasks, we compare feature representations extracted by the backbone ResNet-50, the baseline method CLA, and our feature extractors F_t and F. A-distance and max $J(\mathbf{W})$ [12] are adopted to evaluate feature transferability and discriminability. The results for the

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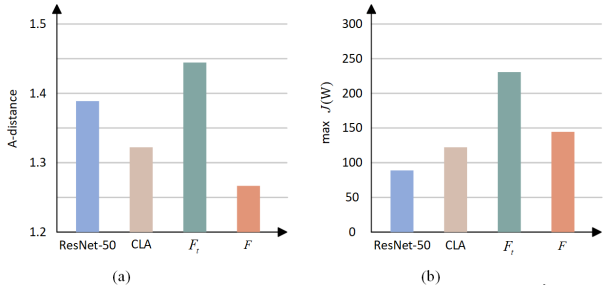


Fig. 12. (a) Transferability and (b) discriminability of four feature representations extracted by ResNet-50, CLA, F_t, and F on A→N task.

A→N task are shown in Fig. 12. Fig. 12(a) illustrates the A-distance of different methods, where a larger A-distance indicates greater domain discrepancy and weaker transferability. Fig. 12(b) presents the max $J(\mathbf{W})$ values, where a higher score suggests better class separability and stronger discriminability. In terms of transferability, the features extracted by CLA exhibit significantly higher transferability than the nonadaptive backbone ResNet-50. This indicates that domain adversarial training improves the ability of the model to generalize across domains. Furthermore, the features learned by our main feature extractor F show stronger transferability than CLA, demonstrating superior domain alignment. Regarding discriminability, the proposed target-specific extractor F_t obtains the most discriminative feature representations, which can generate high-confidence pseudolabels for the unlabeled target domain. In addition, the discriminability of features learned by the main extractor F also surpasses CLA. In summary, our method effectively balances transferability and discriminability, achieves an optimal tradeoff between them, and outperforms the current state-of-the-art PDA method CLA.

V. Conclusion

This article proposes SCAPDA, a novel method for PDA in RSSC. To address the limitations of existing PDA methods that often overlook discriminative learning during adaptation, SCAPDA explores the discriminative structure of the target domain and introduces supervised contrastive learning to synergize adversarial training. In particular, DCM is designed to guide the training of a target-specific classifier, thereby providing high-confidence supervision signals for contrastive learning. Meanwhile, SCLM is proposed to jointly mine domain-invariant and class-specific information across domains, promoting the convergence of adversarial training to a more balanced point. Extensive experiments on three publicly available remote sensing datasets validate the effectiveness of the proposed method, which outperforms existing PDA approaches across multiple adaptation tasks. This study highlights the importance of jointly modeling feature transferability and discriminability in domain adaptation for remote sensing and offers a practical solution for unlabeled scene classification scenarios.

However, the reweighting strategy depends on the accuracy of the estimated weights. When the weights are inaccurate, misalignment between source and target domains may still occur. In addition, adversarial domain alignment is implicit, lacking explicit class-level guidance. In future work, we plan to explore fine-grained class-level alignment strategies to further improve adaptation performance.

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